AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC 2210: OBJECT ORIENTED PROGRAMMING 2

Fall 2025-2026

Section: N

Project Report On

Event Management System

Supervised By

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				**Total =**
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

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1. INTRODUCTION

Event management is an important and growing sector that requires proper planning, coordination, and management of resources. Managing events manually can be time-consuming, error-prone, and inefficient, especially when handling multiple bookings, budgets, venues, and payments. To overcome these problems, an automated Event Management System is necessary.

The Event Management System is a desktop-based application developed using C# and Microsoft SQL Server. This system allows users to book events by selecting event type, venue, services, and payment method. The system dynamically calculates the total budget based on selected services and prevents duplicate bookings for the same venue on the same date.

The system consists of two main modules: User and Admin. Users can register, log in, and submit event booking requests through a user-friendly booking form. On the other hand, the Admin can securely log in to the system to view all bookings, approve or cancel bookings, update budgets, and delete bookings if necessary.

This project aims to reduce manual work, improve accuracy, and provide an efficient way to manage event bookings and administrative tasks. The system follows real-life business logic and provides a reliable solution for small to medium-scale event management organizations.

2. FEATURE LIST

2.1 User 1(Customer)

1. User registration and secure login system
2. Event booking form with multiple input options
3. Selection of event type, venue, and number of guests
4. Service selection using checkboxes (e.g., decoration, catering, photography, sound system)
5. Dynamic budget calculation based on selected services
6. Read-only budget display to prevent manual editing
7. Advance payment option with payment method and transaction details
8. Automatic prevention of duplicate bookings for the same venue on the same date
9. Booking confirmation after successful payment

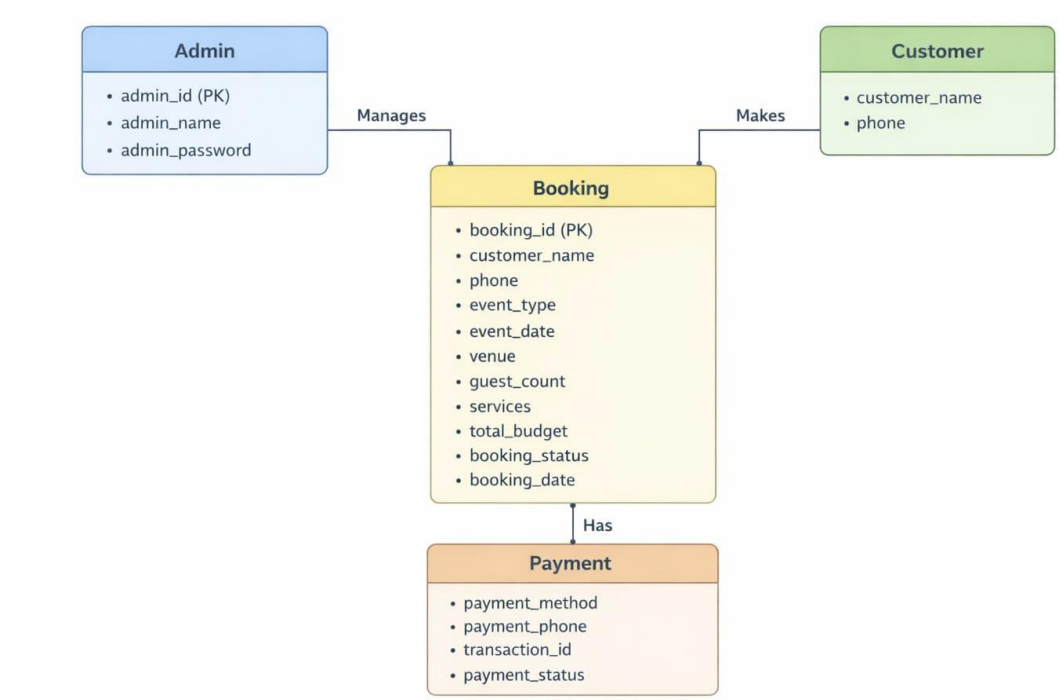
2.2 User 2 (Admin)

1. Secure admin login system
2. Admin dashboard to view all booking records
3. Display of complete booking details including customer information, services, budget, and payment status
4. Ability to approve or cancel bookings
5. Option to edit the total budget of any booking
6. Ability to permanently delete bookings from the system
7. Real-time update of booking status in the database

2.3 System Features

1. Desktop-based application developed using C# (Windows Forms)
2. Microsoft SQL Server used as the backend database
3. Structured database design with proper relationships
4. User-friendly and interactive graphical user interface
5. Error handling and data validation to ensure system reliability

3. ER DIAGRAM



3.1 Normalization

Relation–1 : Admin manages Booking

UNF: Admin_ID, Admin_Name, Admin_Password, Booking_ID, Event_Type, Event_Date, Venue, Guest_Count, Services, Total_Budget, Booking_Status

1NF: Admin_ID, Admin_Name, Admin_Password, Booking_ID, Event_Type, Event_Date, Venue, Guest_Count, Services, Total_Budget, Booking_Status

2NF:
1. Admin_ID (PK), Admin_Name, Admin_Password
2. Booking_ID (PK), Event_Type, Event_Date, Venue, Guest_Count, Services, Total_Budget, Booking_Status
3. Admin_ID (FK), Booking_ID (FK)

Relation–2 : Customer makes Booking

UNF: Customer_ID, Customer_Name, Phone, Booking_ID, Event_Type, Event_Date, Venue, Guest_Count, Services, Total_Budget

1NF: Customer_ID, Customer_Name, Phone, Booking_ID, Event_Type, Event_Date, Venue, Guest_Count, Services, Total_Budget

2NF:
1. Customer_ID (PK), Customer_Name, Phone
2. Booking_ID (PK), Event_Type, Event_Date, Venue, Guest_Count, Services, Total_Budget
3. Customer_ID (FK), Booking_ID (FK)

Relation–3 : Booking has Payment
UNF: Booking_ID, Payment_Method, Payment_Phone, Transaction_ID, Payment_Status

1NF: Booking_ID, Payment_Method, Payment_Phone, Transaction_ID, Payment_Status

2NF:
1. Booking_ID (PK)
2. Payment_ID (PK), Payment_Method, Payment_Phone, Transaction_ID, Payment_Status
3.Booking_ID (FK), Payment_ID (FK)

3.2 Finalization

1. Admin

Admin_ID (PK) ,Admin_Name, Admin_Password

2. Customer

Customer_ID (PK), Customer_Name ,Phone

3. Booking

Booking_ID (PK),Event_Type ,Event_Date,Venue ,Guest_Count,
Services ,Total_Budget ,Booking_Status ,Customer_ID (FK) ,Admin_ID (FK)

4. Payment

Payment_ID (PK),Payment_Method , Payment_Phone, Transaction_ID, Payment_Status
,Booking_ID (FK)

4. UML DIAGRAM

4.1 Use Case Diagram

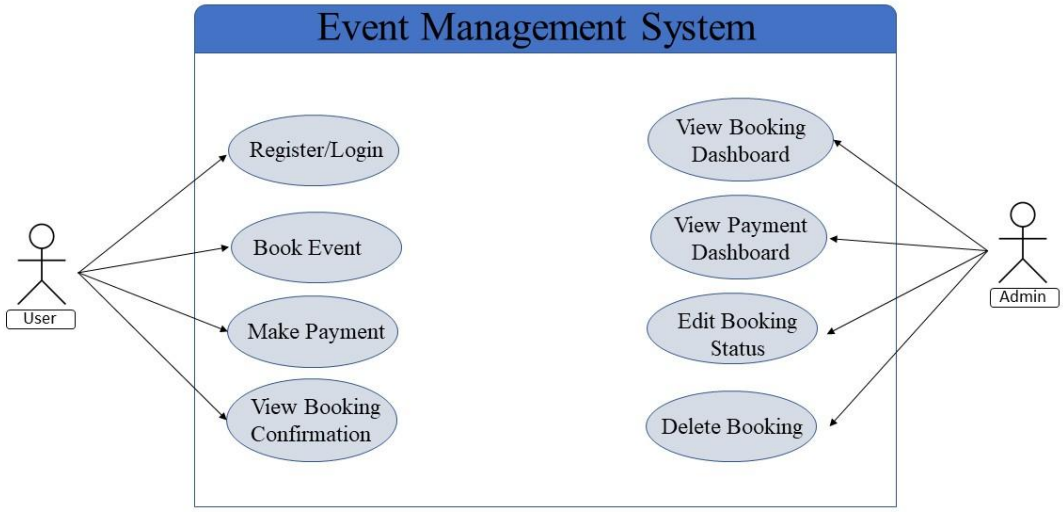


Diagram Description

The Use Case Diagram illustrates the functional requirements of the Event Management System and shows how different actors interact with the system.
In this system, there are two main actors: User and Admin.

Actors Description

User

The User represents customers who want to book events. The user can:

- Register or Login to the system
- Book an event by providing event details
- Make payment for the selected services
- View booking confirmation after successful payment

Admin

The Admin manages the overall system operations. The admin can:

- View booking dashboard
- View payment dashboard
- Edit booking status (Confirm / Cancel)
- Delete booking records if necessary

System Overview

The Event Management System acts as a central platform where users can easily book events and make payments, while admins can monitor, manage, and control bookings efficiently through administrative dashboards.

4.2 Activity Diagram.

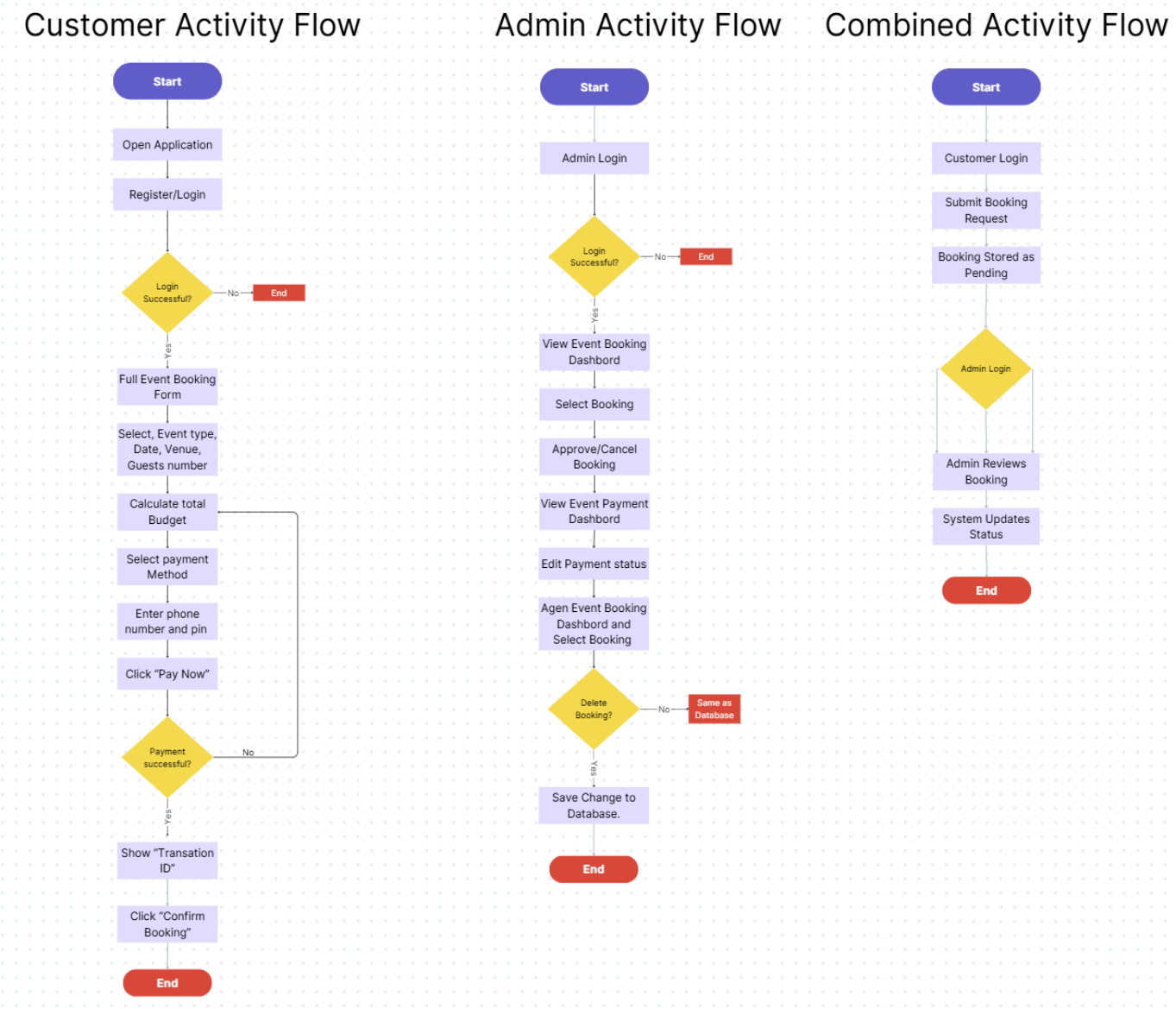


Diagram Description

The Activity Diagram shows the overall workflow of the Event Management System. It explains how customers and admins interact with the system.

Customer Activity Flow

The customer opens the application and registers or logs in.
After successful login, the customer fills up the event booking form by selecting event type, date, venue, number of guests, and services.
The system calculates the total budget.
The customer selects a payment method, makes the payment, and receives a transaction ID.
Finally, the customer confirms the booking.

Admin Activity Flow

The admin logs into the system.
After successful login, the admin views the booking dashboard and selects a booking.
The admin can approve, cancel, edit, or delete bookings and update payment status.
All changes are saved in the database.

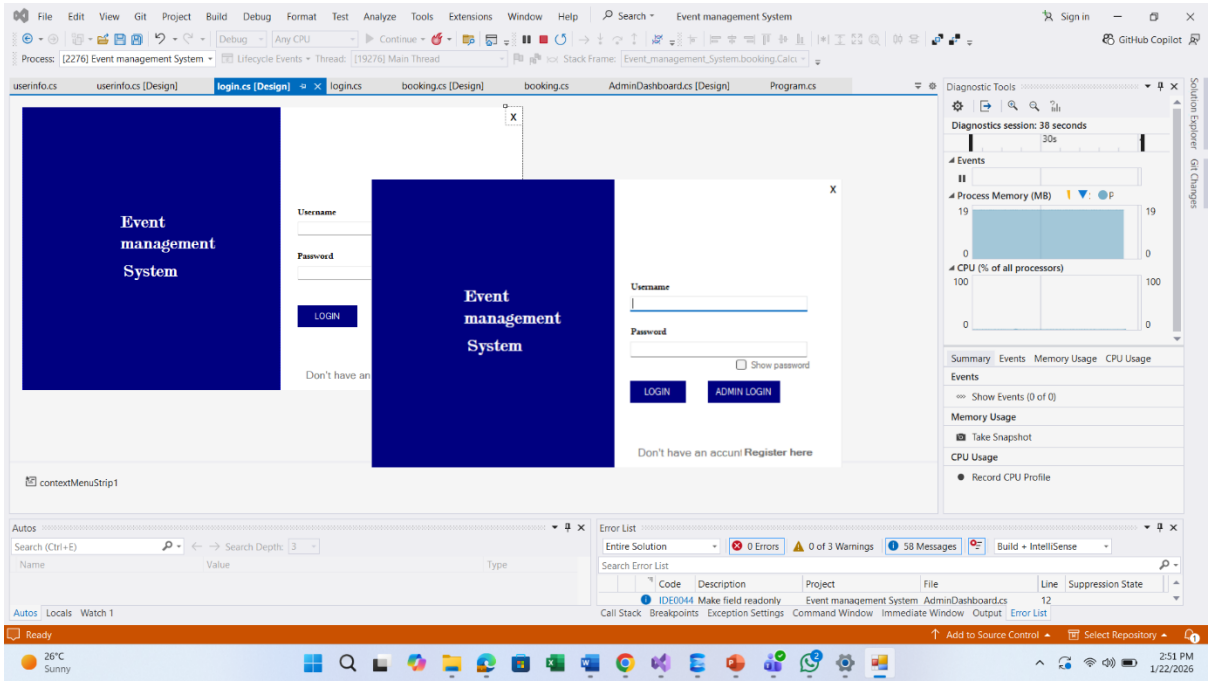
Combined Activity Flow

The customer submits a booking request which is stored as pending.
The admin reviews the booking and updates the booking status.
The system updates the final status and completes the process.

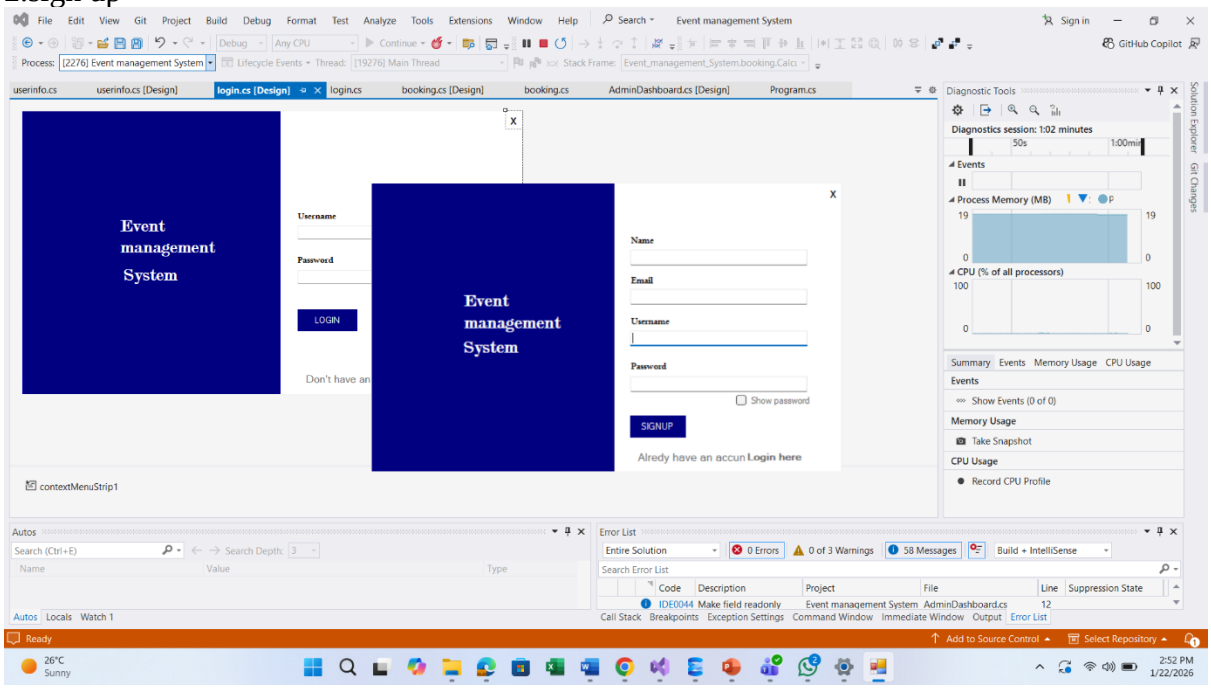
This activity diagram clearly represents the step-by-step process of event booking, payment, and admin management in the system.

5. Project Screenshots

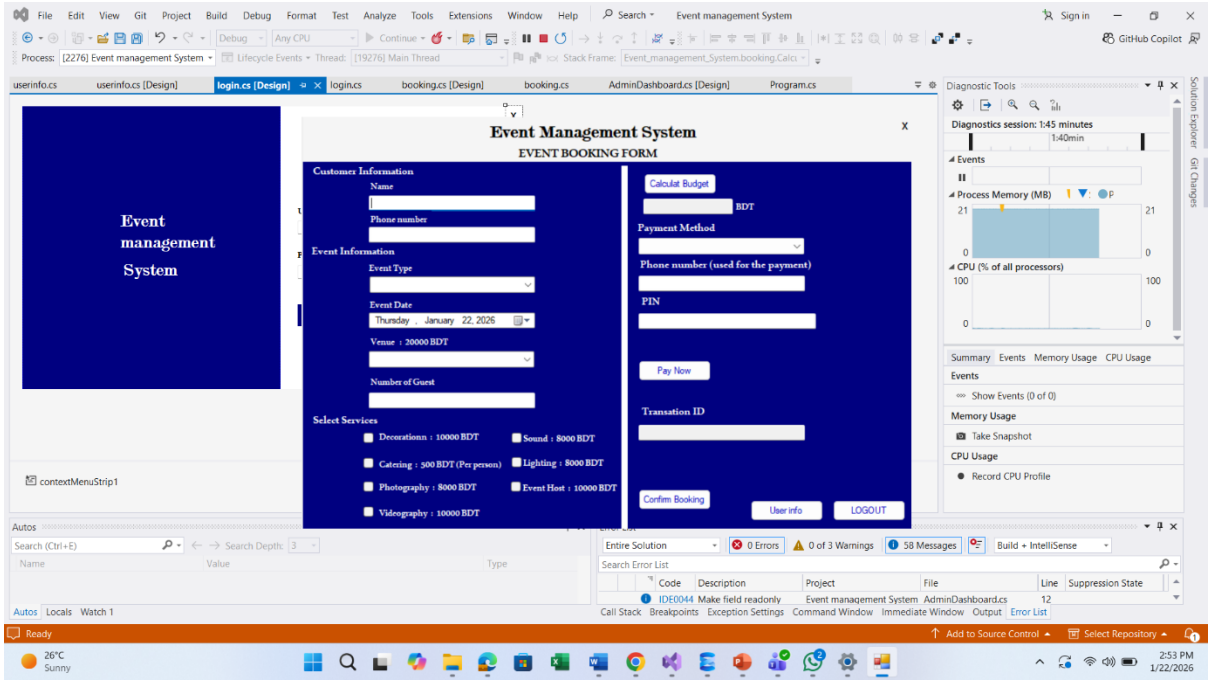
1.



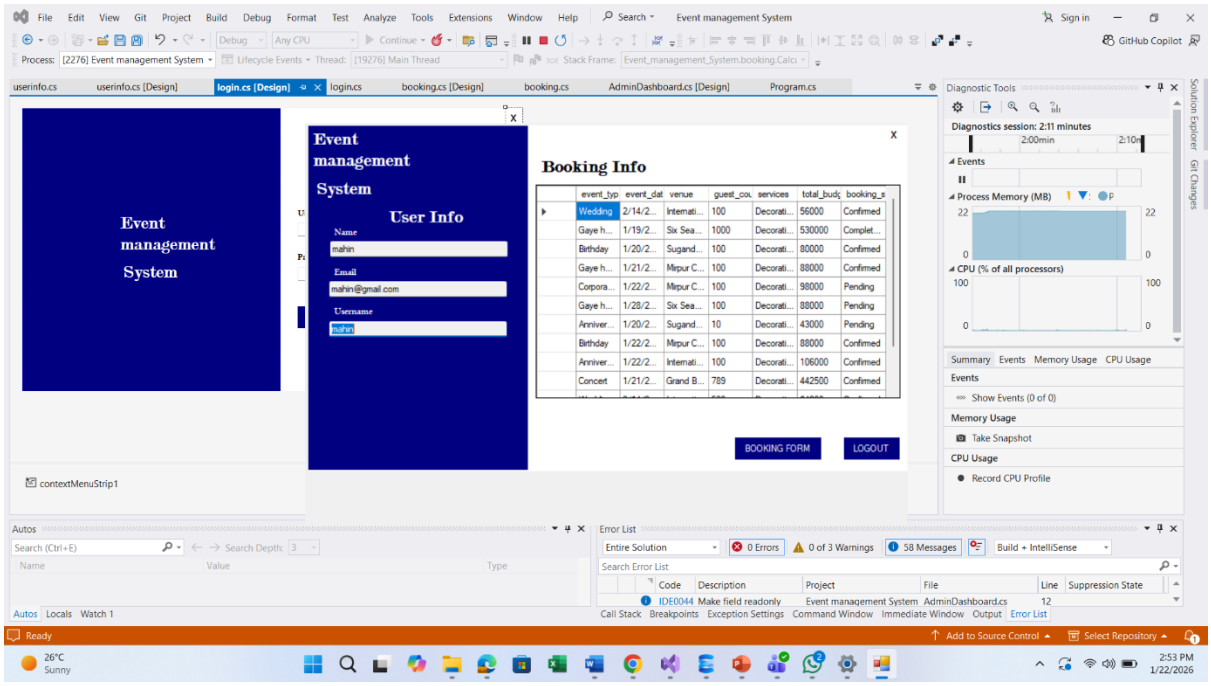
2.sign up



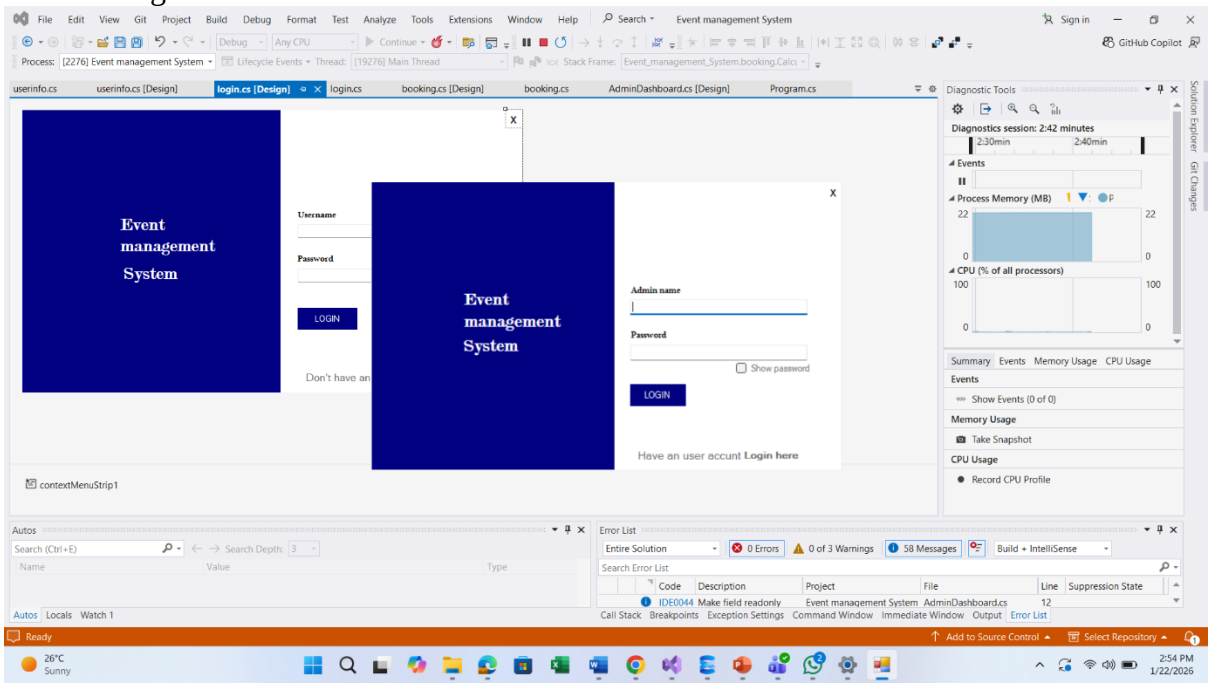
3. Booking form



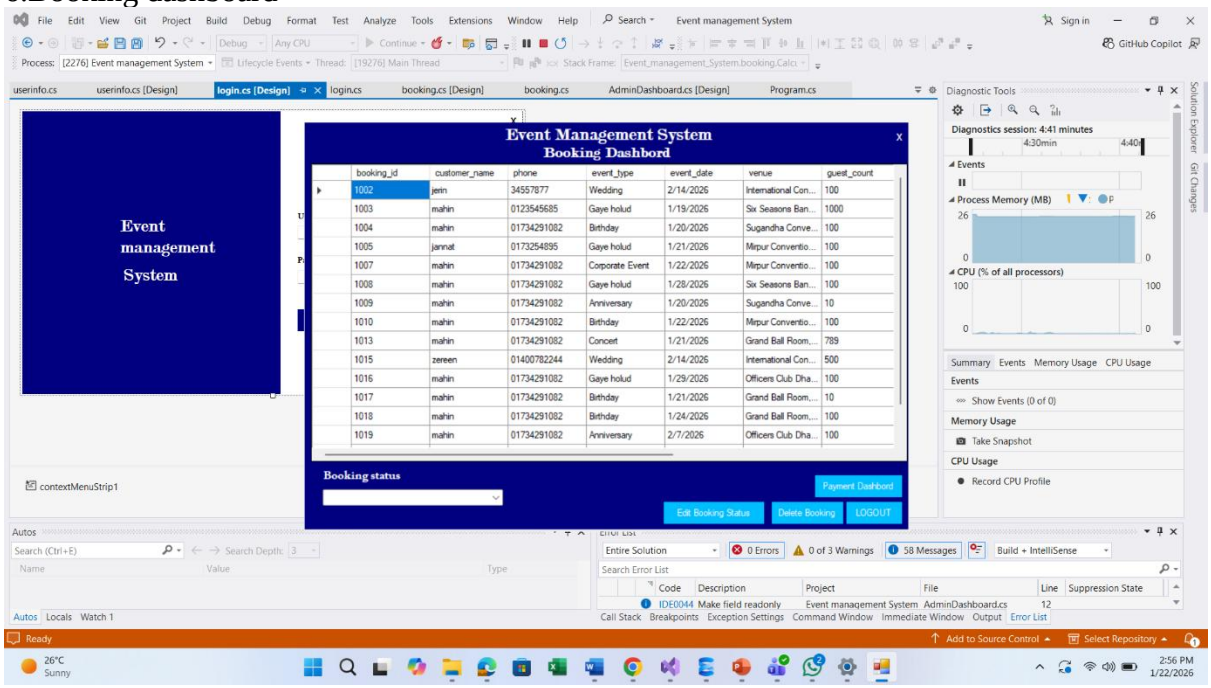
4.User info



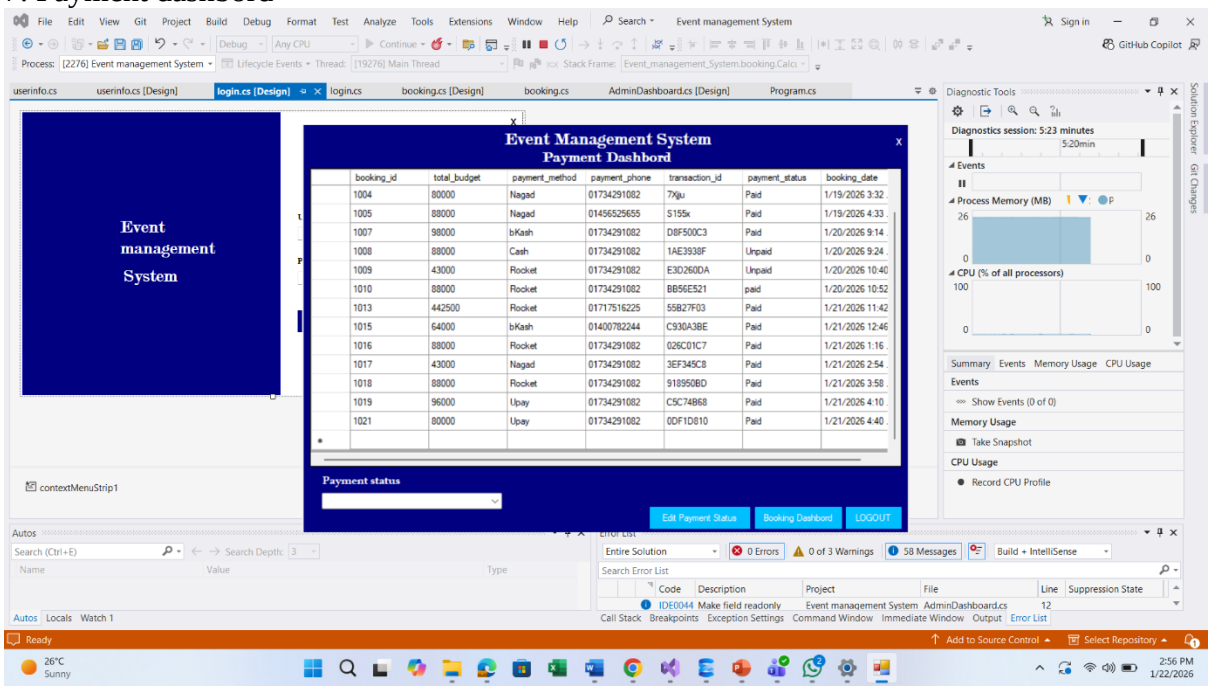
5.Admin login



6.Booking dashboard



7. Payment dashbord



6. CONCLUSION

The Event Management System successfully automates the process of managing event bookings, payments, and administrative tasks. By replacing manual operations with a desktop-based C# application connected to Microsoft SQL Server, the system reduces errors, saves time, and ensures accurate handling of multiple events simultaneously.

The project demonstrates how real-world business logic can be implemented in software to provide efficient solutions for small to medium-scale event management organizations. Users benefit from a seamless booking experience, while admins can easily manage, approve, or cancel bookings and track payments in real time.

Overall, this system improves operational efficiency, minimizes duplication or conflicts in bookings, and provides a reliable, user-friendly interface. It serves as a practical and scalable solution that can be further enhanced with additional features, such as reporting, notifications, or integration with online payment gateways.

Key achievements:

Reduced manual workload and human errors

Ensured real-time updates and secure data management

Provided dynamic budget calculation and payment tracking

Implemented structured database design following proper normalization rules

In conclusion, the system meets its objectives of improving efficiency, accuracy, and reliability in event management, making it a valuable tool for both customers and administrators.